

1) List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

BIOS/POST Error Message	Usually Indicates	BIOS/POST Error Message
1. CMOS Checksum Error	CMOS/BIOS settings are corrupted or the CMOS battery (CR2032) is weak/dead.	1. CMOS Checksum Error
2. Memory Size Decreased / Memory Error	RAM may be loose, incompatible, or faulty.	2. Memory Size Decreased / Memory Error
3. Keyboard Error / Keyboard Not Detected	Keyboard is disconnected, faulty, or the USB/PS/2 port has a problem.	3. Keyboard Error / Keyboard Not Detected
4. No Boot Device Found	HDD/SSD is not detected, boot order is incorrect, or the operating system/boot files are damaged.	4. No Boot Device Found
5. CPU Fan Error	CPU cooling fan is disconnected, not spinning, or the BIOS fan-speed setting is incorrect.	5. CPU Fan Error

2) Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

- **Reseat the RAM**

- Turn off and unplug the PC.
- Remove the RAM modules and reinstall them firmly.
- Make sure the motherboard's locking clips click into place.

- **Test the RAM modules individually**

- If there are two or more RAM sticks, remove all but one.
- Try booting with each module separately and, if necessary, try different RAM slots.
- If one module works and another does not, the faulty RAM module is a likely cause. Intel similarly recommends testing memory modules one at a time for repeated 3-beep memory errors

3) Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

4) You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.